A Q U A P U L S E

Solution Approach & Design System

The Technical 'Why' and the Aesthetic 'Feel' of the AquaPulse Experience

The Core Problem

Urban Rainwater Harvesting (RWH) systems fail not because of a shortage of infrastructure, but because of systematically neglected maintenance. Tanks exist in abundance; the missing ingredient is a digital accountability layer that makes compliance effortless and non-compliance costly.

01 Solution Approach: The Hybrid Compliance Model

AquaPulse addresses urban RWH negligence through a Carrot and Stick digital enforcement strategy — a dual-incentive architecture that simultaneously rewards proactive maintenance while penalising neglect. A third component, The Bridge, removes the friction that makes compliance feel burdensome.

The Carrot — Incentive Layer

Buildings that maintain a 'Healthy' yield score receive Automatic Municipal Tax Rebates, directly credited to their property tax ID. This financial incentive transforms RWH maintenance from a regulatory obligation into a revenue-positive activity for property owners.

The Stick — Enforcement Layer

The moment a building's efficiency score drops below the 35% threshold, an Immediate Digital Defect Notice is auto-generated and delivered directly to the owner's AquaPulse dashboard. This replaces slow manual municipal inspections with real-time, automated enforcement that cannot be gamed or delayed.

The Bridge — Frictionless Compliance

> The Vision AI Logbook allows property owners to prove maintenance completion with a single smartphone photograph. The AI classifies the image, timestamps the event, and appends it to the building's immutable maintenance record — bypassing manual municipal inspections entirely and making compliance the path of least resistance.

02 Design System Specifications

The AquaPulse visual language is engineered for trust, clarity, and performance. Every design decision is purposeful: the interface must communicate urgent data at a glance while projecting the premium feel of enterprise civic software.

Element	Specification	Design Rationale
Typography	**Inter — Sans-Serif**	High legibility for data-heavy dashboards; modern and premium corporate aesthetic.
Design Style	**Glassmorphism / Soft UI**	Translucent metric containers feel lightweight and professional; reduces cognitive load on complex dashboards.
Primary Blue	**#0078D7 (Aqua Pulse)**	Evokes trust, cleanliness, and the clarity of fresh water. Consistent with civic and enterprise software conventions.
Healthy State	**#28a745 — Green**	Industry-standard traffic-light system for instant user comprehension; Healthy = system performing at target.
Warning State	**#ffc107 — Amber**	Signals degraded but non-critical performance; prompts owner action before automatic penalty triggers.
Critical State	**#dc3545 — Red**	Triggers the enforcement layer; unmistakable urgency without requiring the user to read a number.
Micro-Animations	**Pulsing Alerts (CSS @keyframes)**	Increases urgency for critical warnings without intrusive pop-ups or notification overload.

Data Visualisation	**Plotly Grouped Bar Charts + Gauge Meters**	High-contrast rendering ensures readability across Light and Dark modes; gauge meters communicate percentages pre-attentively.

03 Semantic Colour System

The semantic colour scheme is a deliberate implementation of the international traffic-light convention, chosen because it requires zero onboarding. Users comprehend system status within 200 milliseconds of viewing any AquaPulse metric — a critical design requirement for busy building managers and municipal officers reviewing hundreds of records simultaneously.

HEALTHY	WARNING	CRITICAL
#28a745	#ffc107	#dc3545
Yield ≥ 65%	35% ≤ Yield < 65%	Yield < 35% → Notice